

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

Project P10

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

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Researchers discovered that 40 Hz sound pulses synchronize brain gamma waves and trigger immune cells to clear amyloid plaques in Alzheimer's models. Clinical trials now test this in patients, but they use the same fixed schedule: stimulate 40 seconds, rest 20, repeat. When I analyzed EEG from 35 dementia patients, I found half habituate within minutes while the other half stay fully engaged. A fixed protocol cannot account for this.

I built a system that predicts when a patient's brain is about to lose entrainment and adjusts timing accordingly. From the OpenNeuro ds005048 dataset (35 subjects, 7 frontal EEG channels, 250 Hz), I computed phase-amplitude coupling between theta and gamma rhythms as an entrainment biomarker, then trained a causal Temporal Convolutional Network (31,000 parameters) to forecast this biomarker 5 seconds ahead and integrated it into a closed-loop controller.

I swept prediction horizons from 1 to 10 seconds. At 1-2 seconds, simply assuming "nothing changes" works fine. At 5-10 seconds, where a controller actually needs to act, every baseline collapses to negative R-squared while my model holds at 0.25. Replaying the controller on all 35 subjects' real EEG, it targeted 82.6% of low-entrainment windows for stimulation versus 51.7% for reactive control ($p < 0.001$) and reached 91% of the theoretical best performance. Overall alignment rose from 64.5% to 72.1% (Hedges' $g = 1.31$, $p < 0.001$).

Every single patient showed improvement. These results suggest personalized, prediction-driven scheduling could make 40 Hz entrainment therapy more efficient by reducing unnecessary stimulation while preserving therapeutic benefit.